

Strands Labs

[Strands Labs](#) is the experimental arm of Strands Agents - a space for projects that push the boundaries of what AI agents can do. Labs projects explore new domains, validate novel approaches, and move fast. All projects are open source.

While the core Strands Agents SDK provides the foundation for building agents - the agent loop, tool use, model providers, and multi-agent patterns - Labs is where that foundation gets applied to new problem spaces. These are projects that extend agents into areas like physical robotics, simulation-based evaluation, and new programming abstractions. Some Labs projects may eventually graduate into the core SDK or become standalone products; others may remain experimental. The common thread is that they all build on agentic AI in open source and are designed to be used alongside it.

Labs projects are fully functional and published to package repositories, but they move faster and have a wider surface area than the core SDK. Expect more frequent changes, newer integrations, and a focus on enabling research and prototyping alongside production use.

Projects

[Robots](#)

Control physical robots with natural language through Strands Agents. The library provides a policy abstraction layer for vision-language-action models and a hardware abstraction layer for robot control, with tools for camera management, teleoperation, pose storage, and servo communication.

[Robots Sim](#)

Develop and test robot control strategies in simulated environments without physical hardware. Provides two execution modes: full episode execution where the agent specifies a task and the policy runs to completion, and iterative control where the agent observes camera feedback after each batch of steps and adapts its instructions.

[AI Functions](#)

Python functions that behave like standard functions but are evaluated by AI agents. AI Functions enforce correctness through runtime post-conditions rather than prompt engineering alone, enabling developers to build reliable agentic workflows using familiar programming patterns. Supports async execution, parallel workflows, and composing functions into multi-agent systems.

Contributing

Have an experimental idea that pushes AI agents forward? Labs is designed for innovation from across the community. Check the [contributing guide](#) to get started.